

Task 2: Query Understanding Decomposition Task

Agentic Text-to-SQL System (FastAPI + PostgreSQL)

Objective

The goal of this task is to learn how to break a natural language question into structured components before writing SQL. This thinking process is called query decomposition and is the foundation of any AI-powered Text-to-SQL system.

SECTION 1: Simple SELECT Queries (Q1–Q20)

These questions select data from a single table with no filtering or joining.

Q1. List all products

- Intent: List all products
 - Tables: products
 - Columns: All columns
 - Filters: None
 - Joins: None
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Q2. Get all customers

- Intent: List all customers
 - Tables: customers
 - Columns: All columns
 - Filters: None
 - Joins: None
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Q3. Show all orders

- Intent: List all orders
- Tables: orders
- Columns: All columns
- Filters: None
- Joins: None

Q4. List all employees

- Intent: List all employees
- Tables: employees
- Columns: All columns
- Filters: None
- Joins: None

Q5. Get all offices

- Intent: List all offices
- Tables: offices
- Columns: All columns
- Filters: None
- Joins: None

Q6. Show all product lines

- Intent: List all product lines
- Tables: productlines
- Columns: All columns
- Filters: None
- Joins: None

Q7. List all payments

- Intent: List all payments
 - Tables: payments
 - Columns: All columns
 - Filters: None
 - Joins: None
-

Q8. Get product names and prices

- Intent: Show product names and buy prices
 - Tables: products
 - Columns: productName, buyPrice
 - Filters: None
 - Joins: None
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Q9. Get customer names and cities

- Intent: Show customer names and cities
 - Tables: customers
 - Columns: customerName, city
 - Filters: None
 - Joins: None
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Q10. List employee first and last names

- Intent: Show employee names
 - Tables: employees
 - Columns: firstName, lastName
 - Filters: None
 - Joins: None
-

Q11. Get all order dates

- Intent: List order dates
 - Tables: orders
 - Columns: orderDate
 - Filters: None
 - Joins: None
-

Q12. Show product vendor list

- Intent: List unique product vendors
 - Tables: products
 - Columns: productVendor
 - Filters: None
 - Joins: None
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Q13. Get all product codes

- Intent: List product codes
 - Tables: products
 - Columns: productCode
 - Filters: None
 - Joins: None
-

Q14. List all countries from offices

- Intent: List unique office countries
 - Tables: offices
 - Columns: country
 - Filters: None
 - Joins: None
-

Q15. Show all order statuses

- Intent: List unique order statuses
 - Tables: orders
 - Columns: status
 - Filters: None
 - Joins: None
-

Q16. Get all payment amounts

- Intent: List payment amounts
- Tables: payments
- Columns: amount
- Filters: None

- Joins: None
-

Q17. List all job titles

- Intent: List unique job titles
 - Tables: employees
 - Columns: jobTitle
 - Filters: None
 - Joins: None
-

Q18. Get customer phone numbers

- Intent: Show customer phones
 - Tables: customers
 - Columns: customerName, phone
 - Filters: None
 - Joins: None
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Q19. Show product MSRP values

- Intent: Show MSRP per product
 - Tables: products
 - Columns: productName, MSRP
 - Filters: None
 - Joins: None
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Q20. List order numbers

- Intent: List order numbers
 - Tables: orders
 - Columns: orderNumber
 - Filters: None
 - Joins: None
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SECTION 2: JOIN Queries (Q21–Q30)

These questions combine data from two or more tables using JOIN operations.

Q21. Get orders with customer names

- Intent: Show orders alongside customer names
 - Tables: orders, customers
 - Columns: orders.orderNumber, orders.orderDate, customers.customerName
 - Filters: None
 - Joins: orders.customerNumber = customers.customerNumber
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Q22. Get employees with office city

- Intent: Show employees with their office city
 - Tables: employees, offices
 - Columns: employees.employeeNumber, employees.firstName, employees.lastName, offices.city
 - Filters: None
 - Joins: employees.officeCode = offices.officeCode
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Q23. Get payments with customer names

- Intent: Show payments alongside customer names
 - Tables: payments, customers
 - Columns: payments.checkNumber, payments.paymentDate, payments.amount, customers.customerName
 - Filters: None
 - Joins: payments.customerNumber = customers.customerNumber
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Q24. Get order details with product names

- Intent: Show order line details with product names
- Tables: orderdetails, products
- Columns: orderdetails.orderNumber, products.productName, orderdetails.quantityOrdered, orderdetails.priceEach

- Filters: None
 - Joins: orderdetails.productCode = products.productCode
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Q25. Get products with product line description

- Intent: Show products with their product line description
 - Tables: products, productlines
 - Columns: products.productName, productlines.productLine, productlines.textDescription
 - Filters: None
 - Joins: products.productLine = productlines.productLine
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Q26. Get customers with sales rep names

- Intent: Show customers and their assigned sales representatives
 - Tables: customers, employees
 - Columns: customers.customerName, employees.firstName, employees.lastName
 - Filters: None
 - Joins: customers.salesRepEmployeeNumber = employees.employeeNumber
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Q27. Get orders with customer city

- Intent: Show orders with the customer's city
 - Tables: orders, customers
 - Columns: orders.orderNumber, orders.orderDate, customers.customerName, customers.city
 - Filters: None
 - Joins: orders.customerNumber = customers.customerNumber
-

Q28. Get employees and their manager

- Intent: Show each employee alongside their manager's name (self-join)
 - Tables: employees (self-join as e and m)
 - Columns: e.employeeNumber, e.firstName, e.lastName, m.firstName AS managerFirst, m.lastName AS managerLast
 - Filters: None
 - Joins: e.reportsTo = m.employeeNumber
-

Q29. Get order details with product vendor

- Intent: Show order details with the product vendor name
 - Tables: orderdetails, products
 - Columns: orderdetails.orderNumber, products.productVendor, orderdetails.quantityOrdered, orderdetails.priceEach
 - Filters: None
 - Joins: orderdetails.productCode = products.productCode
-

Q30. Get payments with customer country

- Intent: Show payments with the customer's country
 - Tables: payments, customers
 - Columns: payments.amount, customers.customerName, customers.country
 - Filters: None
 - Joins: payments.customerNumber = customers.customerNumber
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SECTION 3: Aggregation Queries (Q31–Q50)

These questions use aggregate functions: COUNT, SUM, AVG, MAX, MIN — often with GROUP BY.

Q31. Count customers per country

- Intent: Count how many customers exist in each country
 - Tables: customers
 - Columns: country, COUNT(customerNumber)
 - Filters: None
 - Joins: None — GROUP BY country
-

Q32. Total payments per customer

- Intent: Sum total payment amounts per customer
- Tables: payments, customers
- Columns: customers.customerName, SUM(payments.amount)
- Filters: None

- Joins: payments.customerNumber = customers.customerNumber — GROUP BY customerName
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Q33. Number of orders per status

- Intent: Count how many orders exist for each status
 - Tables: orders
 - Columns: status, COUNT(orderNumber)
 - Filters: None
 - Joins: None — GROUP BY status
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Q34. Products per product line

- Intent: Count how many products belong to each product line
 - Tables: products
 - Columns: productLine, COUNT(productCode)
 - Filters: None
 - Joins: None — GROUP BY productLine
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Q35. Employees per office

- Intent: Count how many employees work in each office
 - Tables: employees
 - Columns: officeCode, COUNT(employeeNumber)
 - Filters: None
 - Joins: None — GROUP BY officeCode
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Q36. Total stock per product vendor

- Intent: Sum total quantity in stock for each vendor
 - Tables: products
 - Columns: productVendor, SUM(quantityInStock)
 - Filters: None
 - Joins: None — GROUP BY productVendor
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Q37. Average buy price per product line

- Intent: Calculate the average buy price for each product line
 - Tables: products
 - Columns: productLine, AVG(buyPrice)
 - Filters: None
 - Joins: None — GROUP BY productLine
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Q38. Orders per customer

- Intent: Count how many orders each customer has placed
 - Tables: orders, customers
 - Columns: customers.customerName, COUNT(orders.orderNumber)
 - Filters: None
 - Joins: orders.customerNumber = customers.customerNumber — GROUP BY customerName
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Q39. Max MSRP per product line

- Intent: Find the highest MSRP within each product line
 - Tables: products
 - Columns: productLine, MAX(MSRP)
 - Filters: None
 - Joins: None — GROUP BY productLine
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Q40. Min buy price per vendor

- Intent: Find the lowest buy price offered by each vendor
 - Tables: products
 - Columns: productVendor, MIN(buyPrice)
 - Filters: None
 - Joins: None — GROUP BY productVendor
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Q41. Total number of customers

- Intent: Count all customers in the database
- Tables: customers
- Columns: COUNT(customerNumber)
- Filters: None
- Joins: None

Q42. Total number of products

- Intent: Count all products in the database
 - Tables: products
 - Columns: COUNT(productCode)
 - Filters: None
 - Joins: None
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Q43. Total revenue from payments

- Intent: Sum all payment amounts to get total revenue
 - Tables: payments
 - Columns: SUM(amount)
 - Filters: None
 - Joins: None
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Q44. Average product price

- Intent: Calculate the average MSRP across all products
 - Tables: products
 - Columns: AVG(MSRP)
 - Filters: None
 - Joins: None
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Q45. Max payment amount

- Intent: Find the single largest payment made
 - Tables: payments
 - Columns: MAX(amount)
 - Filters: None
 - Joins: None
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Q46. Min payment amount

- Intent: Find the single smallest payment made
- Tables: payments
- Columns: MIN(amount)

- Filters: None
 - Joins: None
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Q47. Count total orders

- Intent: Count all orders in the database
 - Tables: orders
 - Columns: COUNT(orderNumber)
 - Filters: None
 - Joins: None
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Q48. Total quantity in stock

- Intent: Sum the total quantity in stock across all products
 - Tables: products
 - Columns: SUM(quantityInStock)
 - Filters: None
 - Joins: None
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Q49. Average MSRP

- Intent: Calculate the average MSRP across all products
 - Tables: products
 - Columns: AVG(MSRP)
 - Filters: None
 - Joins: None
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Q50. Number of employees

- Intent: Count all employees in the database
 - Tables: employees
 - Columns: COUNT(employeeNumber)
 - Filters: None
 - Joins: None
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